

Elijah Shaw-Rutschman

Staff-Level Software Engineer · Open-Source Systems, Compilers & Realtime Audio · Platform & Infrastructure (Nim, Python, C)

Chicago, IL (US)

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🐙 [elijahr](https://github.com/elijahr) · <https://github.com/elijahr> 🐙 [axiomantic](https://github.com/axiomantic) · <https://github.com/axiomantic>

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SUMMARY

Open-source systems and platform engineer with nearly two decades of experience. Author and maintainer of well-regarded systems libraries in Nim and Python (lock-free concurrent data structures, epoch-based memory reclamation, and realtime audio), a contributor to the Nim compiler's type system and code generation, and author of a library included in the Nim compiler's regression test suite. Spent twelve years at StyleSeat, a Python and Django marketplace, on platform, payments, and reliability, including a Stripe Radar fraud-detection system and a fault-tolerant Redis client library. Earned StyleSeat's highest performance rating ("Exceptional") and was on the formal Staff Engineer promotion track. Chicago-based and open to remote roles across the US.

SKILLS

Languages

Python · Nim · TypeScript · Cython · C · SQL

Backend

Django · Celery · REST APIs

Data & Infrastructure

MySQL · Redis · AWS · Terraform · Docker · CircleCI · GitHub Actions

Systems & Low-Level

Lock-free concurrency · Memory reclamation · Realtime audio / DSP · Low-level protocols · C / Cython interop

Compiler & Language Tooling

Compiler internals · Code generation · libclang · C/C++ header parsing · Cython codegen · Cross-compilation · QEMU multi-arch CI

Payments

Stripe billing · Stripe subscriptions · Stripe Radar · Webhooks · Fraud detection

AI & LLM Tooling

AI-assisted engineering · LLM application development · AI agents · MCP (Model Context Protocol) · AI code review

Practices

Release engineering · Incident response · TDD · Mentoring

WORK EXPERIENCE

Senior Software Engineer II at StyleSeat

Feb 2014 - Jun 2026

Twelve years wearing many hats on a Python/Django marketplace: full-stack product, native mobile, search, payments, reliability, and most recently DevOps and platform engineering. Grew into a senior engineer the company relied on for its hardest, most cross-cutting problems.

- **AI code review:** Built "StyleSeatBot," an org-wide automated code-review bot powered by Claude. It became a required reviewer on every pull request, signing off alongside a human approver, with adoption up to the CTO.
- **AI enablement:** Helped drive the company's move to AI-assisted engineering as a member of the AI-adoption tiger team, running internal talks and workshops on Claude Code and LLM fundamentals.

- **Payments:** Longtime key payments-platform engineer and the team's payments domain expert; major contributor to the Stripe billing and subscriptions build-out (promotion codes, plan and product backfills, coupon-stacking prevention, Stripe Portal, Pro Earnings tab).
- **Fraud detection:** Architected and built an automated fraud-detection system on Stripe Radar.
- **Platform migrations:** Led the pdm-to-uv migration; contributed to org-wide Python 2-to-3 (~900K LOC), Segment-to-Braze, and MySQL 5.7-to-8 migrations.
- **Reliability:** Owned a custom fault-tolerant Redis client library for graceful degradation when Redis was down, and built an S3-backed idempotent Celery task-recovery toolkit after a broker outage failed 351 tasks.
- **Release engineering:** Owned release tooling: a custom CircleCI orb, deploy-gate and hotfix/cherry-pick workflows, and build-failure alerting; contributed to CI/CD and Terraform/ECS.
- **Scale & mentorship:** Among the highest PR-review and code-authoring volume on the team; delivered work across product, platform, payments, and infrastructure codebases, and mentored engineers.

Software Engineer at Switchcam

Aug 2012 - Feb 2014

📍 San Francisco, CA / Chicago, IL

- Full-stack product engineering at a social-video startup, building Python/Django backends and JavaScript frontends across data models, APIs, and UI.

Lead Django Developer at Fulton Works

Feb 2011 - Aug 2012

📍 Chicago, IL

- Led Django development of client web applications at a development studio, owning architecture and full-stack delivery across multiple engagements.

Software Engineer at Hagerty Consulting

May 2009 - Feb 2011

📍 Evanston, IL

- Built internal web applications and data tooling in Python/Django to support consulting engagements.

Software Engineer / IT Engineer at MSDSONline

May 2006 - May 2009

📍 Chicago, IL

- Developed web application features for a SaaS chemical-safety and compliance platform, moving into software engineering from an IT and internal-tooling role.

PROJECTS

nim-lang/Nim (compiler contributor)

Skills: Nim · Compiler internals · Type systems · Code generation

🔗 <https://github.com/nim-lang/Nim/pulls?q=is%3Apr+author%3Aelijahr+is%3Amerged>

Merged fixes to the Nim compiler's type system and C code generator, including cycle detection for recursive concepts, sizeof for typedesc in generic template when clauses, C codegen for lifecycle hooks on generic distinct types, and object initialization for opaque importc fields.

Upstream open-source contributions

Skills: Nim · Python · C · Realtime audio · Open source

🔗 <https://github.com/elijahr>

Merged contributions to language and systems projects, including Nim ecosystem tools (nph formatter, futhark), Python's Black formatter and Cython, and low-level C and realtime-audio libraries (nng, RtAudio, pynng).

lockfreequeues

Skills: Nim · Lock-free · Concurrency

🔗 <https://github.com/elijahr/lockfreequeues>

Lock-free queues for Nim (SPSC, SPMC, MPSC, MPMC; bounded and unbounded). 50+ stars, in the Nimble registry. One of the community libraries in the Nim compiler's regression suite (testament important_packages), built against compiler HEAD in CI to catch regressions.

nim-debra

Skills: Nim · Memory reclamation · Systems

🔗 <https://github.com/elijahr/nim-debra>

Epoch-based memory reclamation (DEBRA+) for lock-free Nim data structures. Nimble package, 15 releases.

python-autopxd2

Skills: Compiler tooling · libclang · Cython · C/C++

🔗 <https://github.com/elijahr/python-autopxd2>

Cython .pxd binding generator for C and C++ headers. As the primary maintainer, rewrote it onto a libclang backend that handles real-world headers. 52+ stars, on PyPI.

run-on-arch-action (major contributor)

Skills: CI/CD · QEMU · Cross-architecture · GitHub Actions

🔗 <https://github.com/uraimo/run-on-arch-action>

GitHub Action to run CI on non-x86 CPUs (ARM, aarch64, s390x, ppc64le, riscv64) via QEMU. Rewrote it to be more maintainable and testable, reviving community contribution. 747 stars.

pynng (co-maintainer)

Skills: Low-level protocols · Messaging · nng · Python

🔗 <https://github.com/codypiersall/pynng>

Python bindings for nanomsg-NG (nng), a low-level messaging and protocol library. 328 stars.

ringbuf

Skills: Realtime audio · DSP · Cython · Lock-free

🔗 <https://github.com/elijahr/ringbuf>

Lock-free SPSC ring buffer for realtime audio in Python and Cython. 22 stars, on PyPI.

axiomantic / spellbook

Skills: AI tooling · MCP · LLM · Developer experience

🔗 <https://github.com/axiomantic/spellbook>

Multi-platform harness for AI coding assistants: skills, hooks, and a shared MCP server across Claude Code, OpenCode, Codex, Gemini CLI, and ForgeCode. Anchor of the axiomantic org.

EDUCATION

Bachelor of Science Computer Science, Minor in Philosophy at
Northeastern Illinois University

Aug 2004 - May 2008